

Chapter 2: Smoothing Covariates

Trick sheet for the exercises

Kernel estimators, Nadaraya–Watson, conditional edf, bandwidths, boundary effects, k-NN

Standing notation (used everywhere below)

Kernel $K \geq 0$, bounded, support in $[-M, M]$, $\int K = 1$, symmetric ($K(-u) = K(u)$). Constants: $\mu_j(K) = \int u^j K(u) du$, $\mu_2(K) = \int u^2 K$, $R(K) = \int K^2$. Density of X : f . $m^{(j)}(z) = \mathbb{E}[Y^j | X = z]f(z)$, $m = m^{(1)}$, $e(z) = \mathbb{E}[Y | X = z] = m(z)/f(z)$, $\sigma^2(z) = \text{Var}(Y | X = z)$. Estimators: $f^{(n)}(x) = \frac{1}{nh} \sum_i K(h^{-1}(x - X_i))$, $m^{(n)}(x) = \frac{1}{nh} \sum_i Y_i K(h^{-1}(x - X_i))$, $e^{(n)} = m^{(n)}/f^{(n)}$. Weights $w_i(x) = K(h^{-1}(x - X_i)) / \sum_j K(h^{-1}(x - X_j))$, so $e^{(n)}(x) = \sum_i w_i Y_i$ and $\sum_i w_i = 1$.

The tricks are grouped by what they do. Section 7 maps each exercise to the tricks it needs.

1 The calculus core: computing means and variances of kernel sums

Trick 1: Change of variables $u = (x - y)/h$ (the Jacobian eats one $1/h$)

For any function g ,

$$\mathbb{E}\left[\frac{1}{h} K\left(\frac{x-X}{h}\right) g(X)\right] = \int \frac{1}{h} K\left(\frac{x-y}{h}\right) g(y) f(y) dy = \int K(u) g(x - hu) f(x - hu) du.$$

Book-keeping rule: *each factor $\frac{1}{h}$ is absorbed by $dy = h du$ only once.* Hence

$$\mathbb{E}\left[K^p\left(\frac{x-X}{h}\right)\right] = h \int K^p(u) f(x - hu) du = O(h) \quad \text{for every } p \geq 1.$$

This is why $\mathbb{E}[(h^{-1}K)^2] \sim f(x)R(K)/h$: squaring leaves one uncanceled $1/h$, which is the source of the $1/(nh)$ variance.

Trick 2: Condition on X to turn regression into density calculations

By the tower property, $\mathbb{E}[Y^j K(\frac{x-X}{h})] = \mathbb{E}[\mathbb{E}[Y^j | X] K(\frac{x-X}{h})]$, so

$$\mathbb{E}\left[\frac{1}{h} Y^j K\left(\frac{x-X}{h}\right)\right] = \int K(u) m^{(j)}(x - hu) du.$$

Every Priestley–Chao computation is the density computation with f replaced by $m^{(j)}$. The same works with any $g(Y)$: replace f by $z \mapsto \mathbb{E}[g(Y) | X = z]f(z)$. It also covers discrete Y (no joint density needed), e.g. $g(Y) = \mathbf{1}(Y \leq t)$ gives $F(t | z)f(z)$.

Trick 3: Second-order Taylor expansion inside the integral

With $f \in C^2$ and $|u| \leq M$ (compact support),

$$f(x - hu) = f(x) - hu f'(x) + \frac{1}{2} h^2 u^2 f''(x) + o(h^2) \quad \text{uniformly in } |u| \leq M.$$

Integrate term-by-term against K . Because the remainder is uniform on the support and K is bounded, $\int K(u) o(h^2) du = o(h^2)$. (Gaussian kernel: not compactly supported, but all moments are finite, so the same works if f'' is bounded and continuous.) If you only have f Lipschitz, stop at first order and get $O(h)$.

Trick 4: Symmetry of K kills all odd moments

K symmetric $\Rightarrow \int u^{2j+1} K(u) du = 0$. Also K^2, K^3 are symmetric, so $\int u K^2(u) du = 0$ too. Consequences:

- the $O(h)$ term in the bias vanishes $\Rightarrow$ bias is $O(h^2)$:

$$\mathbb{E}[f^{(n)}(x)] - f(x) = \frac{h^2}{2} \mu_2(K) f''(x) + o(h^2), \quad \mathbb{E}[m^{(n)}(x)] - m(x) = \frac{h^2}{2} \mu_2(K) m''(x) + o(h^2);$$

- $\int K^2(u)f(x-hu)du = f(x)R(K) + O(h^2)$ (the book writes $O(h)$, which is also true);
- at a boundary (half-kernel) symmetry gives $\int_0^M K = \frac{1}{2}$ but $\int_0^M uK(u)du \neq 0$, which is exactly why boundary bias is $O(1)$ or $O(h)$ instead of $O(h^2)$ (Trick 27).

Trick 5: Variance of an iid kernel average: “the squared mean is always lower order”

For $\xi_i = \frac{1}{h}g(Y_i)K(\frac{x-X_i}{h})$ iid,

$$\text{Var}\left(\frac{1}{n}\sum_i \xi_i\right) = \frac{1}{n}\left(\mathbb{E}\xi_1^2 - (\mathbb{E}\xi_1)^2\right) = \frac{1}{nh}\int K^2(u)G_2(x-hu)du - \frac{1}{n}\left(\int K(u)G_1(x-hu)du\right)^2,$$

where $G_k(z) = \mathbb{E}[g(Y)^k | X=z]f(z)$. The first term is $\frac{G_2(x)R(K)}{nh}(1+o(1))$ and the second is $O(1/n)$, which is negligible since $h \rightarrow 0$. Examples: $\text{Var } f^{(n)}(x) = \frac{f(x)R(K)}{nh} + O(\frac{1}{n})$, $\text{Var } m^{(n)}(x) = \frac{\mathbb{E}[Y^2|X=x]f(x)R(K)}{nh} + O(\frac{1}{n})$.

Trick 6: Bias–variance decomposition + Chebyshev = consistency

$\mathbb{E}[(T - \theta)^2] = \text{Var}(T) + (\mathbb{E}T - \theta)^2$ and $P(|T - \theta| > \varepsilon) \leq \mathbb{E}[(T - \theta)^2]/\varepsilon^2$. With bias $O(h^2)$ and variance $O(1/(nh))$: $h \rightarrow 0$ and $nh \rightarrow \infty$ give $T \xrightarrow{P} \theta$. Fixed h does *not* work: $\mathbb{E}f^{(n)}(x) = (K_h * f)(x)$ with $K_h(u) = h^{-1}K(u/h)$, a smoothed version of f (Example 2.1).

Trick 7: Rate book-keeping (memorise this table)

Quantity	Order	Condition used
bias (interior, symmetric K , C^2)	h^2	Taylor + symmetry
variance	$1/(nh)$	Trick 1 + Trick 5
$\sqrt{nh} \times \text{bias}$	$\sqrt{nh^5}$	$\rightarrow 0$ iff $nh^5 \rightarrow 0$ (undersmoothing)
stochastic error	$O_P((nh)^{-1/2})$	Chebyshev
$f^{(n)} - f$	$O_P(h^2 + (nh)^{-1/2})$	sum of the two
k -dim. covariates	var. $1/(nh^k)$, $h^* \propto n^{-1/(k+4)}$	curse of dimensionality

Remember $O(1/n) - O(1/n)$ is *not* zero, it is $O(1/n)$.

2 Limit-theorem tricks (pointwise CLTs)

Trick 8: Always split into stochastic part + bias part

$$\sqrt{nh}(T_n - \theta) = \underbrace{\sqrt{nh}(T_n - \mathbb{E}T_n)}_{A_n: \text{CLT}} + \underbrace{\sqrt{nh}(\mathbb{E}T_n - \theta)}_{B_n = O(\sqrt{nh^5}) \rightarrow 0}.$$

For ratios, split around $\mathbb{E}[\text{num}]/\mathbb{E}[\text{den}]$ as in Theorem 2.4, or use Trick 9 (faster).

Trick 9: The centre-at-the-target identity for ratio estimators (exact, no Taylor!)

Because the weights sum to one,

$$e^{(n)}(x) - e(x) = \sum_i w_i(Y_i - e(x)) = \frac{1}{f^{(n)}(x)} \cdot \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x-X_i}{h}\right)(Y_i - e(x)).$$

Generally $A_n/B_n - a/b = (A_n - \frac{a}{b}B_n)/B_n$. Now $f^{(n)}(x) \xrightarrow{P} f(x)$ handles the denominator (Slutsky), and the numerator is a *single* iid kernel sum with response $Y - e(x)$. Its bias is $\frac{h^2}{2}\mu_2(K)D''(x)$ with $D(z) = (e(z) - e(x))f(z)$, giving the NW bias

$$\frac{h^2}{2}\mu_2(K)\left(e''(x) + \frac{2e'(x)f'(x)}{f(x)}\right),$$

and its variance gives $\sigma^2(x)R(K)/f(x)$ directly. Works for the cdf ($Y \rightarrow \mathbf{1}(Y \leq t)$), mean excess, etc.

Trick 10: Kernel covariance rule (read off limit covariances instantly)

Let $S_a = (nh)^{-1/2} \sum_i \{K(\frac{x-X_i}{h})g_a(Y_i) - \mathbb{E}[\cdot]\}$. Then (Tricks 1, 2, 5)

$$\text{Cov}(S_a, S_b) \rightarrow R(K) f(x) \mathbb{E}[g_a(Y)g_b(Y) | X = x].$$

Note there is **no** “ $-\mathbb{E}g_a \mathbb{E}g_b$ ” term (it is $O(h)$). The centring appears only after dividing by $f^{(n)}$ with Trick 9: take $g_a(Y) = \tilde{g}_a(Y) - \mathbb{E}[\tilde{g}_a(Y) | X = x]$ and divide by $f(x)^2$:

$$\text{limit covariance} = \frac{R(K)}{f(x)} \text{Cov}(\tilde{g}_a(Y), \tilde{g}_b(Y) | X = x).$$

Examples: NW gives $\sigma^2(x)R(K)/f(x)$; cdf at s, t gives $\frac{R(K)}{f(x)}(F(s \wedge t | x) - F(s | x)F(t | x))$, i.e. a Brownian bridge $B \circ F(\cdot | x)$ times $(R(K)/f(x))^{1/2}$.

Trick 11: Cramér–Wold: joint normality via one scalar array

To get joint normality of several kernel sums (e.g. $f^{(n)}, m^{(n)}$, or $f^{(n)}, m^{(n)}, m_2^{(n)}$), consider

$$Z_{ni} = (nh)^{-1/2} \left\{ K\left(\frac{x-X_i}{h}\right)(\lambda_1 + \lambda_2 Y_i + \lambda_3 Y_i^2) - \mathbb{E}[\cdot] \right\}$$

for arbitrary λ 's, verify Lindeberg for $\sum_i Z_{ni}$, and read the covariance matrix from the coefficient of $\lambda_a \lambda_b$. Then *choose* the λ 's equal to the gradient of the smooth map (e.g. $\lambda_1 = -e(x)/f(x)$, $\lambda_2 = 1/f(x)$ for NW).

Trick 12: Lyapunov with $\alpha = 3$ via Minkowski (the “factor 8” trick)

For $Z = \tilde{Z} - \mathbb{E}\tilde{Z}$, Minkowski gives $\|Z\|_3 \leq \|\tilde{Z}\|_3 + |\mathbb{E}\tilde{Z}| \leq 2\|\tilde{Z}\|_3$ (since L^p norms increase in p), so $\mathbb{E}|Z|^3 \leq 8\mathbb{E}|\tilde{Z}|^3$. With $\tilde{Z}_{ni} = (nh)^{-1/2} K(\cdot) |g(Y_i)|$:

$$\sum_{i=1}^n \mathbb{E}|Z_{ni}|^3 \leq 8n(nh)^{-3/2} h \int K^3(u) G_{|g|^3}(x-hu) du = O((nh)^{-1/2}) \rightarrow 0.$$

Generally $\sum \mathbb{E}|Z_{ni}|^{2+\delta} = O((nh)^{-\delta/2})$. This needs $\mathbb{E}[|g(Y)|^3 | X]$ (or $2+\delta$) finite near x . If you only have fewer moments (e.g. $g = Y^2$ with only $\mathbb{E}[Y^4 | X] < \infty$), check Lindeberg directly instead.

Trick 13: Slutsky, continuous mapping, and “ $o_P(1)$ differences don’t matter”

- $f^{(n)}(x) \xrightarrow{P} f(x) > 0 \Rightarrow 1/f^{(n)}(x) \xrightarrow{P} 1/f(x)$ (continuous mapping, Thm 1.4).
- Ch. 1 Exercise 6: $\|Z_n - W_n\| = o_P(1)$ and $Z_n \xrightarrow{d} Z \Rightarrow W_n \xrightarrow{d} Z$. Use it to replace random denominators by their limits and drop negligible terms.
- Terms like $\sqrt{nh}(e^{(n)} - e)^2 = O_P(\sqrt{nh}(h^4 + (nh)^{-1})) = o_P(1)$ can be dropped.

Trick 14: Finite-dimensional δ -method for smooth functions of kernel sums

If $\sqrt{nh}(\mathbf{T}_n - \boldsymbol{\theta}) \xrightarrow{d} N(0, \Sigma)$ then $\sqrt{nh}(g(\mathbf{T}_n) - g(\boldsymbol{\theta})) \xrightarrow{d} N(0, \nabla g^\top \Sigma \nabla g)$. Typical maps: $g(a, b) = b/a$ with $\nabla g = (-b/a^2, 1/a)$; $g(a, b, c) = c/a - b^2/a^2$ (conditional variance, Ex. 6). The gradient evaluated at (f, m, m_2) is the vector of λ 's in Cramér–Wold.

Trick 15: Asymptotic independence at distinct points: disjoint kernel windows

If $x_1 \neq x_2$ and $\text{supp } K \subseteq [-M, M]$, then for $2Mh < |x_1 - x_2|$, $K(\frac{x_1-z}{h})K(\frac{x_2-z}{h}) = 0$ for all z . So the product term in the covariance vanishes and only the product of the means remains: after the $(nh)^{-1} \cdot n$ scaling it is $h^{-1}O(h)O(h) = O(h)$, and even $O(h^5)$ when responses are centred at $e(x_j)$ (Theorem 2.7). Finite-dimensional limits at different x 's are independent.

Trick 16: Almost sure convergence: truncate, Bernstein, Borel–Cantelli

Split $Y_i = Y_i \mathbf{1}(|Y_i| \leq M_n) + Y_i \mathbf{1}(|Y_i| > M_n)$ with $M_n = \sqrt{nh/\log n}$.

- Bounded part: Bernstein $P(|\cdot| > \varepsilon) \leq 2 \exp\{-C\varepsilon^2 nh/(1 + \varepsilon M_n)\}$, summable $\Rightarrow$ Borel–Cantelli.
- Tail part: $|y| \mathbf{1}(|y| > b) \leq |y|^{2+\delta} b^{-(1+\delta)}$, then Markov and summability of $(\log n/(nh))^{(1+\delta)/2}$.

3 Process-level (functional) tricks**Recipe for any functional CLT in this chapter (Theorem 2.8, Exercises 4, 5, 8, 9)**

1. **Linearise:** the process is (kernel sum in t)/ $f^{(n)}(x)$; since the denominator is a scalar (not a process), handle it by Slutsky/Trick 9. Only the numerator needs pathwise treatment.
2. **Finite-dimensional limits:** Cramér–Wold + Lyapunov + Trick 10 identify the covariance.
3. **Tightness:** bracketing CLT for changing classes (Theorem 1.20) with the envelope and brackets below.
4. **Bias:** $\sup_t |\text{bias}(t)| = O(h^2)$ uniformly; $\sqrt{nh^5} \rightarrow 0$.
5. **Transform:** functional δ -method (Theorem 1.17) with a Hadamard-differentiable map.

Trick 17: Rescale so the kernel process is an empirical process

Put $f_{n,t}(y, z) = h^{-1/2} \mathbf{1}_{[0,t]}(y) K(h^{-1}(z - x))$. Then $\sqrt{n}(\mathbb{P}_n - P)f_{n,t} = \sqrt{nh}(\tilde{X}(t) - \mathbb{E}\tilde{X}(t))$ with $\tilde{X}(t) = \frac{1}{nh} \sum_i \mathbf{1}(Y_i \leq t) K(\cdot)$. This puts you in the setting of Theorem 1.20.

Trick 18: Envelope conditions (1.6)–(1.7): moments of $h^{-1/2}K$

Envelope $F_n = h^{-1/2}K(h^{-1}(X - x))$. By Trick 1: $\mathbb{E}F_n^2 = f(x)R(K) + O(h) = O(1)$. For the Lindeberg-type condition use

$$F_n^2 \mathbf{1}(F_n > c\sqrt{n}) \leq \frac{F_n^3}{c\sqrt{n}} \quad \Rightarrow \quad \mathbb{E}[F_n^2 \mathbf{1}(F_n > c\sqrt{n})] \leq \frac{f(x) \int K^3 + O(h)}{c\sqrt{nh}} \rightarrow 0.$$

Trick 19: Bracketing indicator classes: grid of F -quantiles

Choose $0 = t_0 < \dots < t_k = \infty$ with $F(t_j - | x) - F(t_{j-1} | x) \leq \varepsilon^2/2$; possible with $k \leq \lceil 2/\varepsilon^2 \rceil$ since F is monotone and bounded by 1. Brackets $l_j = h^{-1/2} \mathbf{1}_{[0, t_{j-1}]} K$, $u_j = h^{-1/2} \mathbf{1}_{[0, t_j]} K$ satisfy $\|u_j - l_j\|_{L^2} \leq \varepsilon \|F_n\|_{L^2}$ (Trick 2 with $g = \mathbf{1}(t_{j-1} < Y < t_j)$). Hence

$$\int_0^1 \sqrt{\log N_{[]} d\varepsilon \leq \int_0^1 \sqrt{\log(2/\varepsilon^2 + 1)} d\varepsilon < \infty.$$

If the partition does not depend on n , condition (1.8) is automatically unnecessary. For non-indicator families (e.g. $(y - u)_+$, monotone in u and 1-Lipschitz) build brackets from a grid in u in the same way.

Trick 20: Hadamard derivatives you will need (Lemma 1.10, Prop. 1.2, Ch. 1 Ex. 15)

Map $\phi(F)$	Derivative $\phi'_F(h)$	Conditions
$F \mapsto F^{-1}(\alpha)$ (quantile)	$-h(F^{-1}(\alpha))/f(F^{-1}(\alpha))$	$F \in C^1$ near quantile, $f > 0$; tangential to C
$F \mapsto \varphi \circ F$	$\varphi'(F)h$	φ' bounded, uniformly continuous
$F \mapsto -\log(1 - F)$ on $[0, \tau]$	$h/(1 - F)$	$F(\tau) < 1$ (keeps $1 - F$ away from 0)
$F \mapsto \int_0^\tau Fw$	$\int_0^\tau hw$	linear, w bounded
ratio $L_1(F)/L_2(F)$	$\frac{L_1(h)}{L_2(F)} - \frac{L_1(F)L_2(h)}{L_2(F)^2}$	$L_2(F) \neq 0$
composition	chain rule $\psi'_{\phi(F)} \circ \phi'_F$	Theorem 1.16

Tip: the limit $B \circ F$ is continuous when F is, so “tangentially to C ” is enough.

Trick 21: Brownian bridge/time-change facts

$\text{Cov}(B(s), B(t)) = s \wedge t - st$, so $\text{Var } B(F(t)) = F(t)(1 - F(t))$ and $\text{Var } B(\alpha) = \alpha(1 - \alpha)$. $B(F(t))/(1 - F(t))$

has covariance $\frac{F(s \wedge t)}{1-F(s \wedge t)}$, i.e. it is a Brownian motion run at clock $F/(1-F)$. $\sup_t |B(F(t))| = \sup_u |B(u)|$ for continuous F (Kolmogorov–Smirnov law; 95% quantile ≈ 1.36).

4 Bias, bandwidth and boundary tricks

Trick 22: Minimising $Ah^4 + B/(nh)$

$\frac{d}{dh} : 4Ah^3 - \frac{B}{nh^2} = 0 \Rightarrow h^* = \left(\frac{B}{4A}\right)^{1/5} n^{-1/5}$, and at h^* both terms are of the same order:

$$Ah^{*4} + \frac{B}{nh^*} = \frac{5}{4} B^{4/5} (4A)^{1/5} n^{-4/5}.$$

So nonparametric MSE decays like $n^{-4/5}$, slower than the parametric n^{-1} of MLE. For AMSE of $m^{(n)}$: $A = \frac{1}{4}(m'')^2 \mu_2(K)^2$, $B = \mathbb{E}[Y^2 | X = x]f(x)R(K)$.

Trick 23: Integrating over x : the tower trick

$\int \mathbb{E}[g(Y) | X = x]f(x) dx = \mathbb{E}[g(Y)]$. So $\int \mathbb{E}[Y^2 | X = x]f(x) dx = \mathbb{E}[Y^2]$ (Ex. 2(b)), and the integrated bias term becomes $\int (m''(x))^2 dx$. Assume these integrals are finite.

Trick 24: Squared density estimator = double sum of the self-convolution $K_{(2)}$

$$\int f^{(n)}(x)^2 dx = \frac{1}{n^2 h^2} \sum_{i,j} \int K\left(\frac{x-X_i}{h}\right) K\left(\frac{x-X_j}{h}\right) dx.$$

Substitute $x = X_i + hu$: the integral is $h \int K(u) K\left(u + \frac{X_i - X_j}{h}\right) du$. Then $w = -u$ and symmetry $K(-w) = K(w)$ turn it into $h K_{(2)}\left(\frac{X_i - X_j}{h}\right)$, and $K_{(2)}$ is itself symmetric, so the sign of the argument does not matter.

Trick 25: Leave-one-out: add and subtract the diagonal

$f_{-i}^{(n)}(X_i) = \frac{1}{(n-1)h} \sum_{j \neq i} K\left(\frac{X_i - X_j}{h}\right)$ and $\sum_{j \neq i} = \sum_j - \{j = i\}$, with the diagonal equal to $K(0)$. Hence

$$\frac{2}{n} \sum_i f_{-i}^{(n)}(X_i) = \frac{2}{n(n-1)h} \sum_{i,j} K\left(\frac{X_i - X_j}{h}\right) - \frac{2K(0)}{(n-1)h}.$$

For the Gaussian kernel $K(0) = 1/\sqrt{2\pi}$, which produces the isolated term in Ex. 3(b).

Trick 26: Convolution of densities = density of a sum

If K is the density of U , then $K_{(2)}$ is the density of $U_1 + U_2$ (independent copies). Gaussian: $N(0, 1) * N(0, 1) = N(0, 2)$, so $K_{(2)}(v) = \frac{1}{2\sqrt{\pi}} e^{-v^2/4}$ (no integral needed). Uniform on $[-1, 1]$: triangular on $[-2, 2]$.

Trick 27: Boundary bias: integrate over half the kernel

Support $[0, \infty)$ and $x = 0$: $y = -hu \geq 0$ forces $u \leq 0$, so with $\kappa_1 = \int_0^M uK(u) du$,

$$\mathbb{E}f^{(n)}(0) = \int_0^M K(v)f(hv) dv = \frac{1}{2}f(0) + h\kappa_1 f'(0) + o(h) \quad (\text{bias} \approx -\frac{1}{2}f(0) = O(1)).$$

Reflection $f_R^{(n)}(x) = \frac{1}{nh} \sum_i \{K(\frac{x-X_i}{h}) + K(\frac{x+X_i}{h})\}$ has mean $\int K(u)f(|x-hu|) du$; at 0 it is $f(0) + 2h\kappa_1 f'(0) + o(h)$. The $O(1)$ loss is gone, an $O(h)$ bias remains, and it becomes $O(h^2)$ if $f'(0) = 0$ (“shoulder”).

Watch out: ratio estimators at the boundary (Exercise 7)

- For $F^{(n)}(t | 0)$ both numerator and denominator lose the same half-kernel mass, so the factor $\frac{1}{2}$ *cancels in the ratio*. Expanding numerator $\frac{1}{2}M_t(0) + h\kappa_1 M'_t(0)$ with $M_t(z) = F(t | z)f(z)$ and denominator $\frac{1}{2}f(0) + h\kappa_1 f'(0)$ gives

$$\mathbb{E}\text{-type bias of the ratio} \approx 2h\kappa_1 \partial_z F(t | z)|_{z=0},$$

which is $O(h)$, i.e. non-vanishing *relative to the interior* $O(h^2)$. The unnormalised pieces separately have $O(1)$ bias.

- With a symmetric kernel, exactly at $x = 0$, $K(\frac{0-X_i}{h}) + K(\frac{0+X_i}{h}) = 2K(X_i/h)$: the reflected ratio *coincides* with the naive one. Reflection clearly helps the numerator/denominator (and the density), so be explicit about which object you correct and at which point ($x = 0$ versus $x = ch$).
- An $O(h^2)$ rate at the boundary needs the first-order term to vanish, e.g. $\partial_z[F(t | z)f(z)]_{z=0} = 0$ and $f'(0) = 0$, or simply $\partial_z F(t | z)|_{z=0} = 0$ for the ratio.

Also, for ratios the expectation of a ratio is not the ratio of expectations; compute the bias of the linearised version (Trick 9).

5 The translation dictionary: Chapter 1 $\rightarrow$ Chapter 2**Trick 28: Local = iid with effective sample size $nhf(x)/R(K)$**

A kernel-weighted estimator at x behaves like the Chapter 1 iid estimator computed from the conditional law of Y given $X = x$, with n replaced by $nhf(x)/R(K)$. So to *guess* (and check) any limit in Exercises 5, 6, 8, 9:

Chapter 1 (iid)	Chapter 2 (kernel at x)	Chapter 2 (k-NN at x)
$\sqrt{n}$	$\sqrt{nh}$	$\sqrt{\kappa}$
limit variance v	$v \cdot R(K)/f(x)$, with conditional moments	v (no factor)
$F, \mathbb{E}$	$F(\cdot x), \mathbb{E}[\cdot X = x]$	same
extra condition	$nh \rightarrow \infty, nh^5 \rightarrow 0$	$\kappa \rightarrow \infty, \kappa^{3/2}/n \rightarrow 0$

Examples: $\sqrt{n}(F^{(n)} - F) \xrightarrow{d} B \circ F$ becomes Thm 2.8; empirical quantile variance $\frac{\alpha(1-\alpha)}{f(q_\alpha)^2}$ becomes Ex. 5; mean excess $\frac{e^{(2)}(u) - e(u)^2}{1 - F(u)}$ and cumulative hazard $\frac{F(t)}{1 - F(t)}$ become Ex. 8 and Ex. 9.

Trick 29: Influence-function shortcut for the limiting variance

Write the target as $\theta(x) = \Psi(\text{conditional law at } x)$ and find ψ with $\hat{\theta} - \theta \approx \sum_i w_i \psi(Y_i)$, $\mathbb{E}[\psi(Y) | X = x] = 0$. Then

$$\sqrt{nh}(\hat{\theta} - \theta) \xrightarrow{d} N\left(0, \frac{R(K)}{f(x)} \mathbb{E}[\psi(Y)^2 | X = x]\right).$$

ψ 's: mean $Y - e$; cdf $\mathbf{1}(Y \leq t) - F(t | x)$; quantile $-\frac{\mathbf{1}(Y \leq q_\alpha) - \alpha}{f(q_\alpha | x)}$; variance $(Y - e)^2 - \sigma^2$; mean excess $\frac{((Y - u) - e(u|x))\mathbf{1}(Y > u)}{1 - F(u|x)}$; cumulative hazard $\frac{\mathbf{1}(Y \leq t) - F(t|x)}{1 - F(t|x)}$.

Trick 30: Useful identities for the conditional functionals

- Variance as a weighted sum: $\hat{\sigma}^2 = \sum_i w_i (Y_i - e^{(n)})^2 = \sum_i w_i (Y_i - e(x))^2 - (e^{(n)} - e(x))^2$ (cross term vanishes because $\sum_i w_i (Y_i - e^{(n)}) = 0$).
- Mean excess: $\mathbb{E}[(Y - u)_+ | x] = \int_u^\infty (1 - F(t | x)) dt$, so $e(u | x) = \frac{\int_u^\infty (1 - F(t|x)) dt}{1 - F(u|x)}$ (a ratio of integrals of F).
- Cumulative hazard: $\Lambda(t | x) = -\log(1 - F(t | x))$, so $\Lambda^{(n)}(\cdot | x) = \phi(F^{(n)}(\cdot | x))$ with $\phi(G) = -\log(1 - G)$.
- $F^{(n)}(\cdot | x)$ is a genuine (random) distribution function (non-negative weights summing to 1), so it lies in the domain of the inverse/log maps.

6 Kernel constants (for Exercise 1, AMISE, and practical work)

Kernel	$K(u)$	$\mu_2(K)$	$R(K)$	$K(0)$
Uniform	$\frac{1}{2}\mathbf{1}_{[-1,1]}$	1/3	1/2	1/2
Triangular	$(1 - u)\mathbf{1}_{[-1,1]}$	1/6	2/3	1
Epanechnikov	$\frac{3}{4}(1 - u^2)\mathbf{1}_{[-1,1]}$	1/5	3/5	3/4
Biweight	$\frac{15}{16}(1 - u^2)^2\mathbf{1}_{[-1,1]}$	1/7	5/7	15/16
Gaussian	$\frac{1}{\sqrt{2\pi}}e^{-u^2/2}$	1	$\frac{1}{2\sqrt{\pi}}$	$\frac{1}{\sqrt{2\pi}}$

Checking kernel requirements (Ex. 1): non-negative, bounded, compact support; $\int K = 1$ (e.g. $\frac{3}{4}[u - \frac{u^3}{3}]_{-1}^1 = 1$); $\int uK = 0$ by symmetry (odd integrand); $\int u^2K > 0$; all $\int |u|^m K < \infty$ because of bounded support. Normal-reference rule: for Gaussian K and $f = N(\mu, s^2)$, $\int (f'')^2 = \frac{3}{8\sqrt{\pi}s^5}$, so $h_{\text{AMISE}} = (4/3)^{1/5} s n^{-1/5} \approx 1.06 s n^{-1/5}$.

7 Exercise-by-exercise map

Exercise	Tricks and checkpoints
1	Kernel requirements checklist (Section 6). Symmetry $\Rightarrow$ first moment 0. Check: $\int K = 1$, $\mu_2(K) = 1/5$.
2(a)	Trick 22 with $A = \frac{1}{4}(m'')^2\mu_2(K)^2$, $B = \mathbb{E}[Y^2 X = x]f(x)R(K)$. Only the orders of the $o(h^4)$ and $O(1/n)$ terms matter; drop them in AMSE.
2(b)	Integrate AMSE over x and use the tower trick: $\int \mathbb{E}[Y^2 x]f(x)dx = \mathbb{E}[Y^2]$.
2(c)	Plug h^* back: both pieces $\asymp n^{-4/5}$ (Trick 22); MLE MSE $\asymp n^{-1}$ (variance $I(\theta)^{-1}/n$, bias $o(n^{-1/2})$).
3(a)	Expand the square as a double sum, substitute $x = X_i + hu$, then symmetry $\Rightarrow K_{(2)}$.
3(b)	$K_{(2)}$ for Gaussian = $N(0, 2)$ density; leave-one-out with add/subtract diagonal; $K(0) = 1/\sqrt{2\pi}$ gives $\frac{2}{\sqrt{2\pi}h(n-1)}$.
4	Functional recipe: rescaled empirical process, envelope $h^{-1/2}K$ ($\mathbb{E}F_n^2 = f(x)R(K) + O(h)$), $F_n^2\mathbf{1}(F_n > c\sqrt{n}) \leq F_n^3/(c\sqrt{n})$, quantile-grid brackets, entropy integral finite, partition free of $n \Rightarrow$ (1.8) not needed, scalar denominator by Slutsky, covariance via Trick 10.
5	Thm 2.8 + inverse-map Hadamard derivative $-h(q)/f(q x) +$ functional δ -method. Limit = $-\sqrt{R(K)/f(x)}B(\alpha)/f(q_\alpha(x x))$; $\text{Var } B(\alpha) = \alpha(1 - \alpha)$. Need $F(\cdot x) \in C^1$ with positive density on a neighbourhood of $q_\alpha(x)$ (Lemma 1.10, 1(i)).
6	Weighted-sum identity for $\hat{\sigma}^2$; drop $(e^{(n)} - e)^2$ in the CLT; influence $\psi = (Y - e)^2 - \sigma^2$ gives $\tau^2(x) = \frac{R(K)}{f(x)}(\mathbb{E}[(Y - e(x))^4 X = x] - \sigma^4(x))$. Bias: apply Trick 9 to response $(Y - e(x))^2$ with $g(z) = \sigma^2(z) + (e(z) - e(x))^2 - \sigma^2(x)$; get $\frac{h^2}{2}\mu_2(K)[(\sigma^2)'' + 2(e')^2 + 2f'(\sigma^2)'/f]$, plus an $O(1/(nh))$ term $-\sigma^2R(K)/(nhf)$ from $-\mathbb{E}(e^{(n)} - e)^2$. Alternatively: δ -method with $g(a, b, c) = c/a - b^2/a^2$ and Cramér–Wold with $\lambda_1 + \lambda_2 Y + \lambda_3 Y^2$. Lindeberg (moments!).
7	Trick 27 and the boundary warning box: half-kernel integrals, $\int_0^1 K = \frac{1}{2}$, $\kappa_1 = \int_0^1 uK \neq 0$; for the ratio the $\frac{1}{2}$ cancels, leaving $O(h)$; reflection $\Rightarrow$ even extension; $O(h^2)$ needs vanishing first derivative at 0.
8	Write $\hat{e}(u x) = \sum_i w_i(Y_i - u)_+ / \sum_i w_i\mathbf{1}(Y_i > u)$. Pointwise: Theorem 2.6 + ratio δ -method $\Rightarrow$ variance $\frac{R(K)}{f(x)} \frac{e^{(2)}(u x) - e(u x)^2}{1 - F(u x)}$. Functional on $[0, \tau]$ with $F(\tau x) < 1$: joint process CLT for numerator (brackets via monotone, Lipschitz $(y - u)_+$; moment for envelope $ Y $) and denominator, then ratio map (denominator bounded away from 0) + chain rule.
9	$\Lambda^{(n)}(\cdot x) = -\log(1 - F^{(n)}(\cdot x))$ on $[0, \tau]$, $F(\tau x) < 1$; derivative $h/(1 - F)$ (Ch. 1 Ex. 15); Thm 2.8 + functional δ -method. Limit $\sqrt{R(K)/f(x)} \frac{B(F(t x))}{1 - F(t x)}$ with covariance $\frac{R(K)}{f(x)} \frac{F(s \wedge t x)}{1 - F(s \wedge t x)}$.

8 Practical exercises: quick tips

- **Bandwidth scale:** $h \propto n^{-1/5}$. Start from the normal reference $1.06 \hat{s} n^{-1/5}$, or from R's default `bw.nrd0`, which is $0.9 \min(\hat{s}, \text{IQR}/1.34) n^{-1/5}$. Then cross-validate, and always plot $\hat{J}(h)$ or $\hat{R}(h)$ to see a clean minimum.
- **Vectorise the CV criterion** with the double-sum formula of Ex. 3(b): build the matrix $D_{ij} = (X_i - X_j)/h$ once. It is $O(n^2)$ memory, so for large n (e.g. 9613) loop in blocks.
- **PIT the covariate:** replace X by $F_X^{(n)}(X) \in (0, 1)$ so one global bandwidth is sensible; then $f(x) \approx 1$ in variance formulas.
- **Confidence bands for $F^{(n)}(\cdot | x)$:** pointwise $F^{(n)} \pm 1.96 \sqrt{R(K)F^{(n)}(1 - F^{(n)})/(nh\hat{f}(x))}$; uniform (KS)

$F^{(n)} \pm 1.36\sqrt{R(K)/(nh\hat{f}(x))}$. Gaussian kernel: $R(K) = 1/(2\sqrt{\pi})$.

- **Where kernels fail:** heavy tails (Pareto) give undersmoothed, noisy tails and a boundary at the threshold (use reflection or log-transform); the uniform mixture has jumps in f at 0, 1, 2, where the bias is $O(1)$ within h of the jump (the C^2 assumption fails). The MLE wins when the parametric model is right.
- **k-NN:** in 1D, $\hat{f}(x) = \kappa/(2nR_\kappa(x))$ with $R_\kappa(x)$ the distance to the κ -th neighbour; small κ gives a rough estimate, large κ oversmooths. Normalisation $\sqrt{\kappa}$ and no $R(K)/f(x)$ factor.